

10,000-Policy Experiment Pipeline

Six verification steps

Generate

10,000 LRF policies · Small / Medium / Large



Convert

LRF → TRF · Algorithm 1 · Ordering 4



Verify Step 1 — Semantic Fidelity

10,000 policies · 73.12 M evaluations · 10.7 h



Verify Step 2 — Anomaly Detection

7,000 trials · 14 configurations



Verify Step 3 — Scalability & Depth

$n = 5 \rightarrow 400$ · depth $d = 4$



Verify Step 4 — Ordering Comparison

12 orderings · $n \in \{50, 100, 200, 400\}$



Verify Step 5 — ClassBench-ng

8 rulesets · 32 cases · vs FDD / HiCuts



Verify Step 6 — Cython Kernel

5 trials · compiled native C

Verified Results

- ✓ **Fidelity = 100%**
73,120,887 evaluations, FN = 0 (95% UCI $\leq 4.10 \times 10^{-8}$)
- ✓ **Recall = Precision = 100%**
20,868 TP, 0 FP, 0 FN
- ✓ **d = 4 constant**
all $n = 5 \rightarrow 400$ and all 32 ClassBench-ng cases
- ✓ **Cython 40×-41× speedup**
97-177 ns/packet (39× at $n = 25$)
- ✓ **Theorems 1-3, Propositions 1-4**
confirmed